IIT MADRAS / BSDA4001

Data Science & AI Lab
Research Project Proposals

Four deep-learning research directions for team discussion

This briefing compares four promising, non-LLM-reliability project ideas: CrisisGraph, TerraWatch, GeoShield, and EduTrace. Each proposal includes a trainable ML core, public research datasets, credible baselines, explicit hypotheses, and a practical evaluation plan.

Selection philosophy: publication-oriented rigor, meaningful end-to-end training, realistic student-team execution, and measurable benefits over established methods.

Executive recommendation

Priority	Project	Best reason to choose
1	CrisisGraph	Most ambitious cross-disciplinary project: flood segmentation plus infrastructure connectivity.
2	TerraWatch	Excellent research/feasibility tradeoff for satellite change detection and robustness.
3	GeoShield	Clean supervised learning benchmark, ideal for sensor fusion and ablation studies.
4	EduTrace	Strong education alternative centered on trainable sequential and graph models.

Prepared for team discussion | 11 October 2026

At a glance | Decision matrix

Dimension	CrisisGraph	TerraWatch	GeoShield	EduTrace
Task	Flooded road/building mapping + graph analysis	Bi-temporal change segmentation	Multisensor landslide segmentation	Predicting next learner response
Core DL	Multitask CNN / transformer + optional topology-aware loss	Siamese CNN / transformer + consistency learning	Dual encoders + reliability-aware fusion	GRU / attention / concept graph
Primary data	SpaceNet 8	S2Looking / LEVIR-CD+	Landslide4Sense	EdNet / ASSISTments
Primary metric	Class F1 / IoU + road connectivity	Change IoU / F1	Landslide F1 / IoU / PR-AUC	AUC / log loss / calibration
Execution risk	Medium–high	Medium	Low–medium	Low–medium
Special caution	Connectivity truth vs. actual passability	Labels may represent building changes, not all environmental changes	Geographic metadata may be restricted	Accuracy does not prove improved learning

What makes the work research-grade?

• Pose a testable hypothesis, not merely a product feature.

• Reproduce a credible benchmark baseline before proposing novel components.

• Use leakage-resistant geographic/event/learner splits, multiple seeds, and confidence intervals when feasible.

• Show relevant ablations: remove temporal inputs, attention, modalities, graph terms, or time features.

• Evaluate out-of-distribution performance and failure patterns, not just best-case accuracy.

Ranking caveat

The ordering is an editorial recommendation, not a quantitative assessment of acceptance probability. Novelty must be validated with a focused current literature review before claiming a publishable contribution.

01 / CrisisGraph

Topology-aware flood impact and infrastructure accessibility assessment

DOMAIN | Geospatial AI | Computer vision | Disaster response

Problem statement

Flooding can interrupt transportation links and damage buildings. Conventional flood mapping often optimizes pixel segmentation, yet a small missed road segment can dramatically change a connectivity estimate. Develop a deep-learning system that maps flood-affected infrastructure from satellite imagery and evaluates whether its predictions preserve transport-network connectivity.

Research hypothesis

Adding topology-aware objectives or graph-based refinement to strong vision baselines will improve infrastructure-level connectivity estimates, potentially even where pixel-level IoU changes only modestly.

Datasets and access

• SpaceNet 8: pre/post-event satellite imagery and labels for buildings and flooded/non-flooded roads. Official challenge and download details: SpaceNet 8.

• Baselines: official baseline and winning solution code: SpaceNetChallenge/SpaceNet8.

Models and comparison baselines

• Baseline A: U-Net/ResNet segmentation of roads, buildings, and flood classes.

• Baseline B: Siamese U-Net with paired pre/post-event imagery.

• Baseline C: a stronger encoder/segmentation architecture such as HRNet or SegFormer, budget permitting.

Proposed deep-learning methodology

Use shared or paired encoders for pre/post-event inputs and train task-specific heads for roads, buildings, and flood attribution. Extract predicted road centerlines into graphs; compare basic postprocessing against a topology-aware correction or differentiable connectivity surrogate. Keep graph-network modeling optional until basic extraction is reliable.

Experimental design and ablations

• Compare post-event-only versus paired-input models.

• Add/remove the connectivity-sensitive component at matched training budgets.

• Hold out distinct events/geographic areas where the benchmark split permits.

• Audit graph errors at intersections, narrow roads, bridges and occluded segments.

Evaluation metrics

Building/road IoU and F1, flooded-class precision/recall, Average Path Length Similarity (APLS) where suitable, road-edge precision/recall, and connectivity preservation.

Practical risks and safeguards

• Graph construction from segmentation masks is brittle; standardize and validate topology extraction.

• Satellite flood labels do not prove roads are traversable. Describe output as estimated accessibility, not safe evacuation guidance.

• Limited geographic diversity may constrain generalization claims.

Expected deliverable

Benchmarked segmentation model, connectivity analysis pipeline, ablations, geospatial results viewer, and research-style report.

02 / TerraWatch

Generalizable deep learning for multi-temporal satellite change detection

DOMAIN | Remote sensing | Computer vision | Representation learning

Problem statement

Multi-temporal satellite image pairs reveal changes to built and natural environments, but illumination, viewpoint differences, seasonal effects, and misregistration can produce false positives. Develop a change-detection model that better isolates annotated changes and remains robust under acquisition and geographic shifts.

Research hypothesis

Consistency regularization or self-supervised temporal feature learning can improve change-mask robustness under controlled perturbations and unseen-domain conditions compared with standard Siamese segmentation.

Datasets and access

• S2Looking: large-scale labeled bi-temporal change dataset; official repository and download instructions: S2Looking dataset.

• LEVIR-CD+: second building-change benchmark in the same dataset repository.

• Label caveat: these benchmarks emphasize annotated changes (often buildings); do not imply they directly label forest loss or every kind of ecological change.

Models and comparison baselines

• Baseline A: absolute image difference + thresholding (simple non-DL check).

• Baseline B: Siamese U-Net / early-fusion U-Net.

• Baseline C: a published attention or transformer-based change detection model.

Proposed deep-learning methodology

Train a dual-stream image encoder with a decoder producing pixel-wise change probabilities. Explore paired-view augmentations, alignment-tolerant feature matching, and cross-time consistency losses. Validate each addition against matched-parameter or matched-training baselines.

Experimental design and ablations

• Compare early fusion vs. Siamese feature fusion.

• Ablate the consistency objective, augmentations, and pretraining strategy.

• Measure robustness to photometric shifts and modest image misalignment.

• Test cross-dataset transfer, clearly documenting differing annotation definitions.

Evaluation metrics

Change IoU, F1/Dice, precision/recall, false positive rate, per-scene scores, and degradation under perturbation.

Practical risks and safeguards

• Cross-dataset labels are not perfectly equivalent; avoid attributing all transfer gaps to domain shift alone.

• Geographic split integrity must be verified to avoid overlapping tiles.

• Artificial perturbations are stress tests, not substitutes for real deployment conditions.

Expected deliverable

Reproducible satellite change-detection benchmark, robustness analysis, trained model, example visualizations, and manuscript-ready ablations.

03 / GeoShield

Reliability-aware multimodal deep learning for landslide segmentation

DOMAIN | Multimodal learning | Remote sensing | Natural hazards

Problem statement

Landslide mapping benefits from optical appearance and terrain conditions, but individual sensors may be noisy or unavailable. Investigate whether a trainable fusion mechanism can use multispectral and terrain cues more effectively than simply stacking input channels.

Research hypothesis

A dual-encoder architecture with learned modality weighting and modality-dropout training will maintain better landslide detection performance under sensor degradation than conventional all-channel fusion.

Datasets and access

• Landslide4Sense 2022: 3,799 train / 245 validation / 800 test image patches, each 128x128 with 14 channels (12 Sentinel-2 multispectral bands, slope, elevation). Official dataset and baseline.

• Important label note: the official repository states that competition mask files are accessible in the training portion; plan reproducible labeled train/validation splits rather than assuming public test labels.

Models and comparison baselines

• Baseline A: official 14-channel U-Net.

• Baseline B: ResU-Net or DeepLabV3+ trained on all channels.

• Baseline C: straightforward two-stream concatenation without attention.

Proposed deep-learning methodology

Separate an optical encoder from a terrain encoder, fuse features using learned gates or cross-attention, and decode a binary landslide mask. Apply synthetic modality dropout/noise during training and test whether fusion weights provide reliable performance gains.

Experimental design and ablations

• Optical-only, terrain-only, and all-channel baselines.

• Ablate gating, dual encoders, and modality dropout.

• Stress-test missing/noisy modalities using identical validation samples.

• Use grouped geographic evaluation only where trustworthy location metadata exists.

Evaluation metrics

Landslide pixel F1, IoU, precision, recall, PR-AUC, and performance loss under controlled sensor degradation.

Practical risks and safeguards

• Class imbalance requires appropriate loss/threshold tuning.

• The official release does not disclose detailed geographic coordinates; do not promise rigorous geographic holdout without additional data.

• Synthetic sensor corruptions may not replicate real cloud cover or acquisition artifacts.

Expected deliverable

Reproducible multisensor segmentation model, controlled robustness benchmark, channel-ablation analysis, inference visualization, and full experimental report.

04 / EduTrace

Temporal and graph-based student knowledge tracing

DOMAIN | Educational data mining | Sequential deep learning | Graph models

Problem statement

Adaptive learning software benefits from estimating how a learner’s mastery evolves over time. Existing knowledge-tracing models may struggle with sparse learner histories, forgetting, and concept relationships. Develop a model for predicting future student responses using temporal and concept-aware representations.

Research hypothesis

Joint modeling of time gaps and concept dependencies will improve prediction and calibration for short-history learners compared with recurrent and attention-only baselines.

Datasets and access

• EdNet: 131,441,538 interactions from 784,309 learners, with multiple levels of recorded behavior. Dataset overview: EdNet paper and dataset; source: riiid/ednet.

• ASSISTments: smaller learner interaction datasets; access via the ASSISTments data portal.

• Reusable baselines: pyKT includes implementations and preprocessing support.

Models and comparison baselines

• Baseline A: Deep Knowledge Tracing (DKT; GRU/LSTM).

• Baseline B: Self-Attentive Knowledge Tracing (SAKT).

• Baseline C: Attentive Knowledge Tracing (AKT) or simpleKT.

Proposed deep-learning methodology

Construct learner sequences of question ID, skill/concept ID, correctness, and time gap where available. Train a time-aware sequence model and incorporate a concept graph through embeddings or a small GNN. Focus evaluation on predictive generalization, not unverified educational effectiveness.

Experimental design and ablations

• Learner-disjoint evaluation and short-history learner cohorts.

• Ablate elapsed-time information, concept graph, and attention.

• Inspect calibration across easy/hard items and history lengths.

• Compare dataset transfers only when concept mappings and label definitions are valid.

Evaluation metrics

Next-response ROC-AUC, accuracy, log loss, Brier score, expected calibration error, and cold-start subgroup performance.

Practical risks and safeguards

- Concept-graph quality may dominate improvements; justify graph construction.
- Sequential data must be chronologically ordered; avoid target/future-information leakage.
- Better prediction does not prove better learning outcomes.
- EdNet has a non-commercial research license; review before redistributing.

Expected deliverable

Trained sequential knowledge-tracing framework, cold-start/forgetting ablations, reproducible benchmark, and optional learner analytics demo.

Suggested execution plan

Phase	Focus	Exit criterion
1 | Scope	Choose task and review 8-12 recent method papers	Frozen hypothesis, dataset access and evaluation split
2 | Baselines	Set up preprocessing and run benchmark methods	Reproducible baseline table
3 | Proposed model	Implement one justified research contribution	Controlled comparison against baselines
4 | Scientific validation	Ablations, leakage checks, OOD tests, seeds	Robust result tables and error analysis
5 | Reporting	Figures, methodology, limitations, demo	Team presentation and paper-style report

Practical team recommendation

Primary: CrisisGraph if the team is prepared for more engineering and a novel downstream infrastructure task. Balanced alternative: TerraWatch. Best scoped supervised deep-learning research: GeoShield. Best non-geospatial direction: EduTrace.

Official links and starter references

- SpaceNet 8 challenge
- SpaceNet 8 challenge methods / code
- S2Looking and LEVIR-CD+ datasets
- Landslide4Sense dataset, U-Net baseline and paper reference
- EdNet overview
- EdNet source repository
- pyKT baselines
- ASSISTments data portal

Research integrity note: These are proposed hypotheses, not established performance claims. Verify dataset licenses, train/test availability, label definitions and the current research literature before final submission.